August 16, 2025

[2]:
```python
import random
import torch
from torch.utils import data
```

1

[3]:
```python
def synthetic_data(w, b, num_examples):
        x = torch.normal(0, 1, (num_examples, len(w)))
        y = torch.matmul(x, w) + b
        y += torch.normal(0, 0.01, y.shape)
        return x, y.reshape((-1, 1))

true_w = torch.tensor([2, -3.4])
true_b = 4.2
features, labels = synthetic_data(true_w, true_b, 1000)
```

2

[4]:
```python
batch_size = 10

def load_array(data_arrays, batch_size, is_train=True):
        dataset = data.TensorDataset(*data_arrays)
        return data.DataLoader(dataset, batch_size, shuffle=is_train)

data_iter = load_array((features, labels), batch_size)
```

3

[5]:
```python
from torch import nn

net = nn.Sequential(nn.Linear(2, 1))

net[0].weight.data.normal_(0, 0.01)
net[0].bias.data.fill_(0)
```

```python
loss = nn.MSELoss()

trainer = torch.optim.SGD(net.parameters(), lr=0.03)
```

4

[16]:
```python
num_epochs = 10

for epoch in range(num_epochs):
        for x, y in data_iter:
                l = loss(net(x), y)
                trainer.zero_grad()
                l.backward()
                trainer.step()
        l = loss(net(features), labels)
        print(f'epoch: {epoch + 1}, loss: {l:f}')
```

```
epoch: 1, loss: 0.000103
epoch: 2, loss: 0.000102
epoch: 3, loss: 0.000102
epoch: 4, loss: 0.000102
epoch: 5, loss: 0.000103
epoch: 6, loss: 0.000102
epoch: 7, loss: 0.000102
epoch: 8, loss: 0.000102
epoch: 9, loss: 0.000102
epoch: 10, loss: 0.000102
```